

FUZAIL AHMED

3D Artist, Ai Engineer and Software/Game Developer

Contact

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Education

Bahria University E8
Bachelor of Computer Science
2022 - 2026

ParkTurk Maarif International
A-lvls (ics)
2020 - 2022

Army Burn Hall College Abbott
O-lvls
2018 - 2020

Summary

3D Modeler, Environment Designer, and CGI Specialist with a strong Computer Science foundation and hands-on experience building high-fidelity visual assets, interactive simulations, and intelligent software pipelines. Proven track record delivering commercial-grade 3D product designs, precise texturing, and cinematic lighting for diverse freelance projects and on contract for PEAK company. Experienced in executing end-to-end CGI production pipelines, including designing and animating dynamic 3D garments within Clo3D and Blender, seamlessly compositing them into live footage, and polishing with advanced motion graphics in After Effects and sound design in Premiere Pro.
Backed by a technical background in applied AI and software engineering, with a portfolio featuring AXIS (a high-fidelity, physics-based surgical simulation application providing real-time operational feedback), a production-ready RAG pipeline translating Gen-Z slang to standard English, and a deep learning pneumonia detection system achieving 90-95% accuracy. Currently completing a Bachelor's degree in Computer Science, actively leveraging technical programming expertise to bridge the gap between cutting-edge 3D graphics pipelines and cloud-native AI deployments.

Skills

Ai , Software and Hardware:

- Deep Learning & Computer Vision
- Generative AI & LLM Architectures
- Retrieval-Augmented Generation (RAG) Pipelines
- NLP Data Pipeline Training
- Interactive Architecture & Physics-Based Simulation
- Real-Time Feedback Loop Engineering
- Full-Stack Web & Software Development
- Custom PCB Design Layout

3D Based:

- 3D Product Modeling
- Environment Design
- Texturing & Look Development
- Map Baking & UV Unwrapping
- Geometry Optimization
- Studio & Cinematic Lighting
- Digital Fashion & Cloth Simulation
- CGI Footage Integration & Compositing
- Motion Graphics Design
- Video Editing & Sound Design (SFX)

Work Experience

Skills

Software and Ai:

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1. Technical & Software Engineering Experience

Lead Simulation Developer | AXIS (Anatomy Exploration and Interactive System)

Jan 2024 - Present

- Developed a high-fidelity, physics-based medical simulation application engineered for surgical procedure practice.
- Programmed real-time feedback systems to analyze and respond dynamically to operational actions during simulated procedures.
- Bridged complex software logic with interactive 3D environments to ensure highly accurate physical feedback.

AI Engineer & Software Developer | Independent Tech Projects

Jun 2022 - Jan 2024

- Developed and optimized a deep learning Pneumonia Detection AI system, achieving a verified 90–95% diagnostic accuracy rate.
- Built and deployed a production-grade Retrieval-Augmented Generation (RAG) translation pipeline to convert Gen-Z slang into standard English.
- Programmed, tested, and shipped multiple hardware-software integration initiatives, web applications, and independent software pipelines

2 . Professional 3D & CGI Experience

3D Product Modeler (Contract) | PEAK Company

Jan 2024 - Present

- Engineered high-fidelity 3D product models tailored for commercial asset lines.
- Executed advanced texturing, look development, and realistic material creation using Substance Painter and Marmoset Toolbag.
- Configured precise studio lighting and rendering pipelines to showcase products with maximum visual accuracy.

Freelance 3D Modeler & Environment Designer | Self-Employed

Jan 2024 - Present

- Conceptualized and constructed immersive 3D environments and high-fidelity assets utilizing Blender and Unreal Engine.
- Collaborated directly with clients to translate creative briefs into production-ready 3D models.
- Managed the complete asset pipeline, including geometry optimization, UV unwrapping, map baking, and engine integration.

CGI & VFX Pipeline Specialist | Project-Based / Freelance

Jan 2024 - Present

- Executed an end-to-end CGI production pipeline to seamlessly composite a dynamic clothing lineup into live-action video footage.
- Modeled, simulated, and animated high-fidelity 3D garments utilizing Clo3D and Blender.
- Supervised post-production workflows, tracking and adding motion graphics in After Effects (AE) and mastering sound design (SFX) in Premiere Pro (PP).

PROJECTS

3D Oriented Projects

Motifz
2025 July

Motifz – End-to-End CGI Advertisement Production
CGI & VFX Pipeline Specialist

Jan 2024 - Present

- Conceptualized and executed a complete CGI production pipeline to integrate a commercial clothing lineup into live-action footage.
- Tailored high-fidelity 3D product designs, executed advanced texturing, and configured studio lighting setups.
- Modeled, simulated, and animated complex, dynamic garments utilizing Clo3D and Blender.
- Supervised post-production workflows, tracking and compositing assets with advanced motion graphics in After Effects and mastering sound design (SFX) in Premiere Pro.

Commercial
24 hrs
2026 July

Heydar Aliyev Center – Architectural & Environment Design

3D & Environment Designer | 24hrs - modeling

- Recreated the iconic architectural landmark as a high-fidelity 3D environment, focusing on its signature fluid, organic surfaces and complex geometric curves.
- Optimized intricate topology to ensure smooth curvature rendering while maintaining clean geometric flow and efficient polygon counts.
- Developed realistic architectural materials and executed atmospheric environmental lighting to capture accurate real-world light distribution and shadows.

PEAK Company (contract)
2026 - Present

Topographic Route Visualizations – PEAK Company

3D Modeling & Digital Fabrication Specialist

- Translated spatial GPS mapping data and Strava routes into high-fidelity, accurate 3D topographic terrain models.
- Engineered production-ready meshes optimized specifically for additive manufacturing, ensuring watertight geometry, correct non-manifold fixes, and structural base stabilization.
- Sculpted and refined complex landscape features, clean geometric borders, and extruded track paths optimized for clear definition when translated into physical 3D prints.

Portfolio
3-4 hrs - modelling
2 hrs - Texturing

J. Oud – Luxury Product Visualization

3D Artist & Material Specialist

- Sculpted and rendered a premium 3D product model of the luxury perfume bottle, emphasizing high-fidelity realism and elegant aesthetics.
- Configured complex glass and liquid shader networks to accurately simulate thick-walled glass refraction, internal fluid volume, and metallic gold accents.
- Designed a dedicated studio lighting setup to highlight the product's silhouettes, textures, and premium branding for high-end portfolio presentation.

Portfolio
6-7 hrs - modelling
3 ½ hrs - Texturing

HP Omen Encoder – Hard Surface Product Design

3D Product Modeler

- Developed a photorealistic, high-fidelity 3D asset of the mechanical keyboard, prioritizing millimeter-precision hard-surface modeling.
- Executed advanced PBR texturing and look development, accurately replicating anodized brushed aluminum, matte plastic keycaps, and braided cable textures.
- Managed the full asset pipeline, including high-to-low poly map baking, UV unwrapping, and setting up a professional studio lighting array for a commercial-grade product showcase.

PROJECTS

AI, Software and Hardware

University
2026 - Present

AXIS (Anatomy Exploration and Interactive System) – Final Year Project

- Lead Developer and Programmer

- Engineered a high-fidelity, physics-based medical simulation application designed for immersive surgical procedure practice.
- Developed core simulation mechanics and interactive 3D environments using C++ and Unreal Engine 5.
- Programmed advanced, real-time feedback loops to dynamically analyze and respond to precise operational actions during simulated procedures.
- Bridged complex software engineering logic with interactive graphics to deliver realistic physical feedback rather than static asset rotation.

University
2025 - 6th sem

Deep Learning Pneumonia Detection AI System

- AI Engineer & Developer

- Designed and trained a robust computer vision pipeline optimized for medical imaging analysis.
- Implemented deep learning classification models to accurately identify anomalies in radiographic data.
- Achieved and verified a strict diagnostic accuracy rate of 90–95% during testing phases.

University
2026 - 6th sem

Gen-Z Slang Translator

- AI Pipeline & Backend Developer

- Built and deployed a production-grade Retrieval-Augmented Generation (RAG) translation system to convert Gen-Z slang into standard English.
- Developed a high-performance backend architecture using Python and FastAPI to manage translation queries.
- Structured, curated, and processed a custom Natural Language Processing (NLP) dataset containing over 1,000+ entries for targeted model training.

Commercial
2025 - Present

Commercial-Grade Ambient Lighting Kit (in Development)

- Founder and Hardware & Firmware Developer

- Engineered an ambient lighting kit for PC monitors from the ground up, seamlessly bridging physical electronics with desktop software.
- Programmed custom firmware and managed device configurations for ESP32-S3 microcontrollers.
- Designed custom PCB layouts and managed strict power constraints to ensure optimal hardware performance and stability.
- Integrated the physical hardware ecosystem directly with SignalRGB for real-time screen-syncing capabilities.

University
2024 - 4th sem

Motion Detection & Smart Traffic Light System

- Hardware & Embedded Systems Developer

- Engineered an automated, wireless traffic management infrastructure designed to dynamically optimize signal timings based on real-time vehicle and pedestrian presence.
- Programmed low-latency embedded firmware to process active motion sensor inputs and orchestrate seamless wireless communication between distributed intersection nodes.
- Developed adaptive scheduling algorithms to minimize traffic congestion, clear gridlocks, and process real-time feedback loops for shifting road demands.
- Designed the physical hardware architecture, managing precise power distribution, sensor calibration matrices, and durable prototype enclosures for synchronized system deployment.